

(Long)LLMLingua | Effectively Deliver Information to LLMs via **Prompt Compression**

aka.ms/LLMLingua

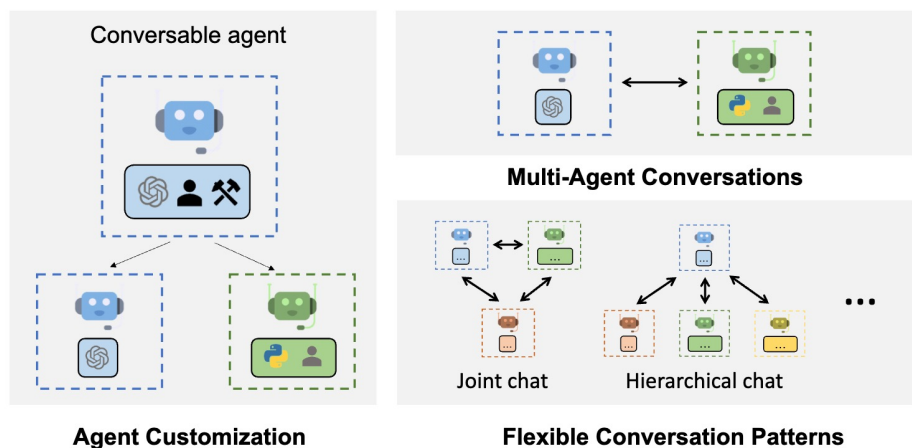
Huiqiang Jiang (MSRA SH)

On behalf of **LLMLingua v-teams**

Joint work from colleagues across DKI, MSAI, M365 teams

Jan. 2024

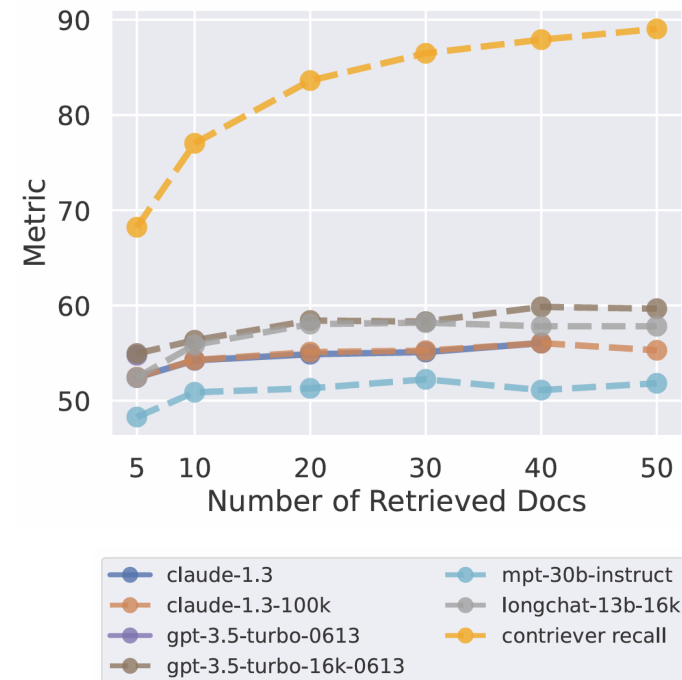
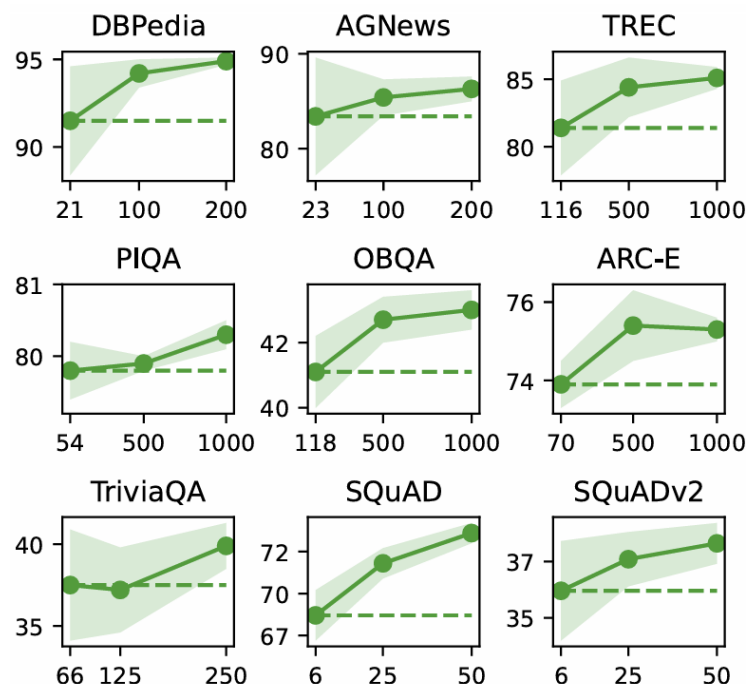
Prompts Become Increasingly Longer



Significant Token Requirements (>10k) in Multi-Turn Agents such as AutoGen^[1], GPTs, Copilots, and Tools.

Adding more demonstrations generally enhances performance.^[2]

More tokens yield a larger recall in RAG.^[3]



However, this requires higher costs, and

* [1] AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation.

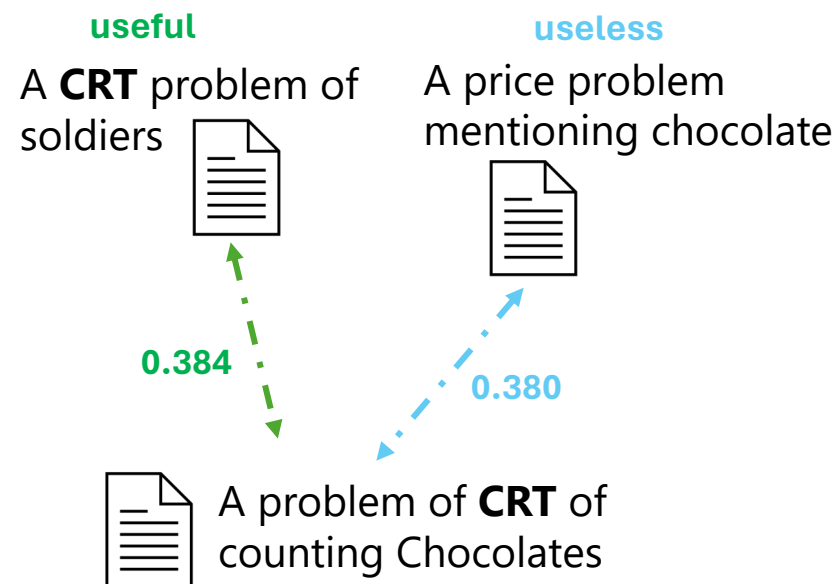
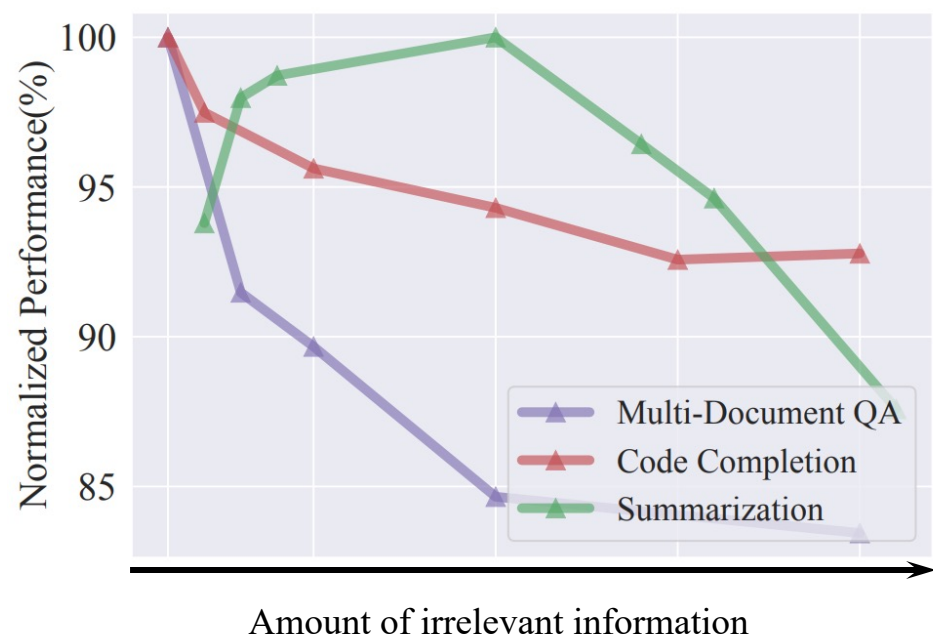
[2] Structured Prompting: Scaling In-Context Learning to 1,000 Examples.

[3] Lost in the Middle: How Language Models Use Long Contexts.

Longer Prompt Tend to Contain More Noise

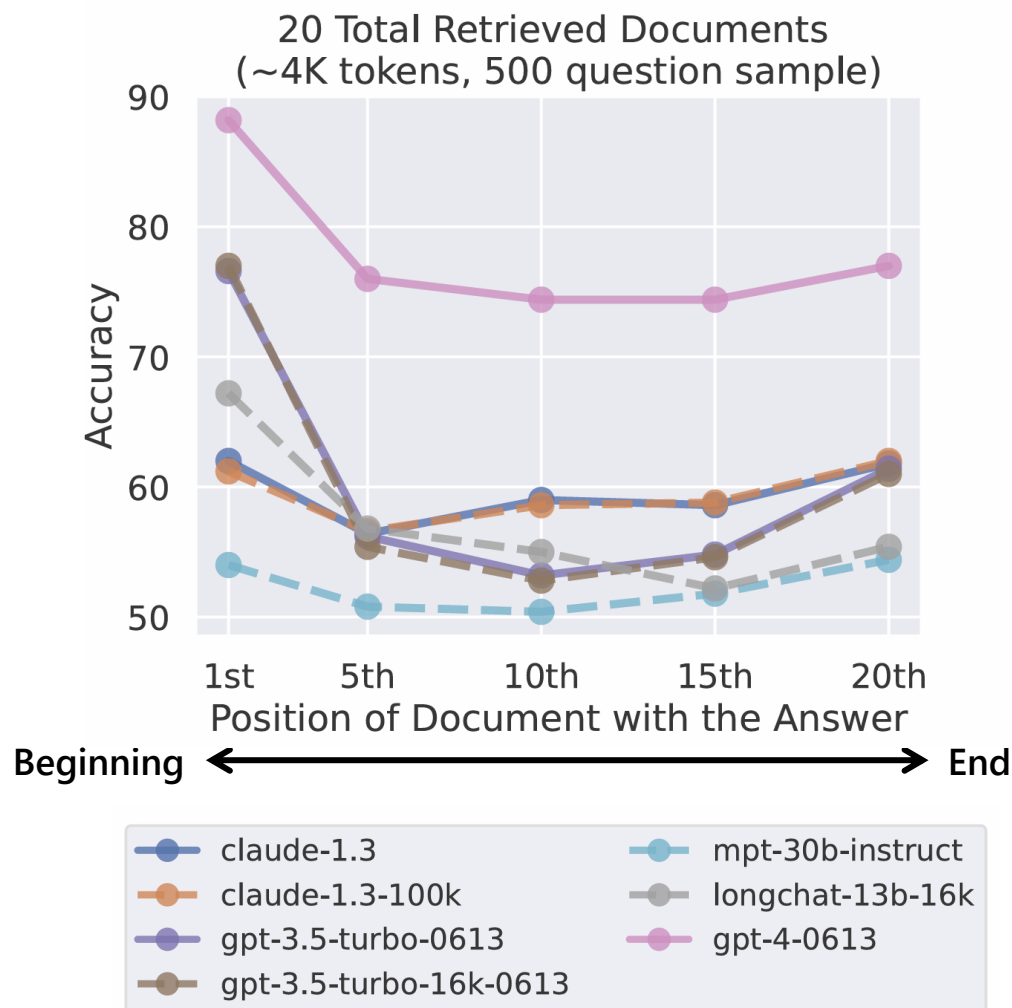
Task may relate to only part of details in raw prompt. LLMs can be easily distracted by irrelevant context^[4]. We should Improve key information density!

Embedding similarity base retrievers struggle to distinguish question-related fine-grained semantic information.

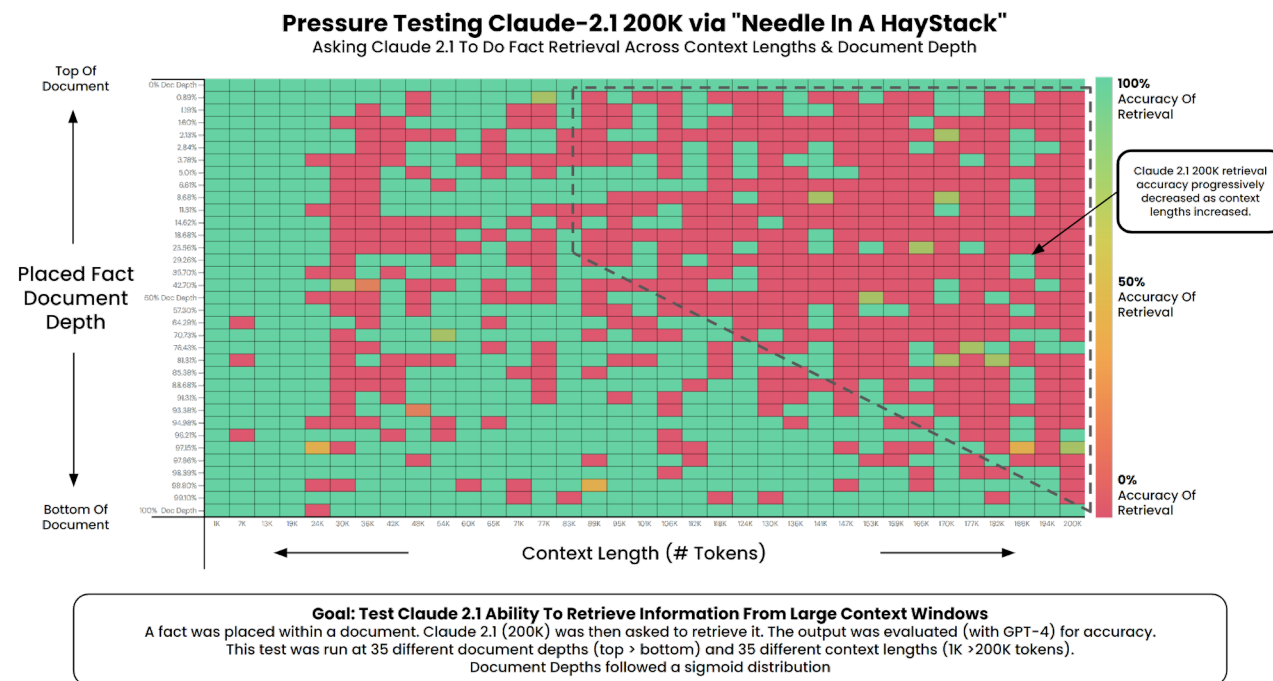


* [4] Large language models can be easily distracted by irrelevant context.

LLMs Have Position Attentive Priority



The accuracy in different position^[3].

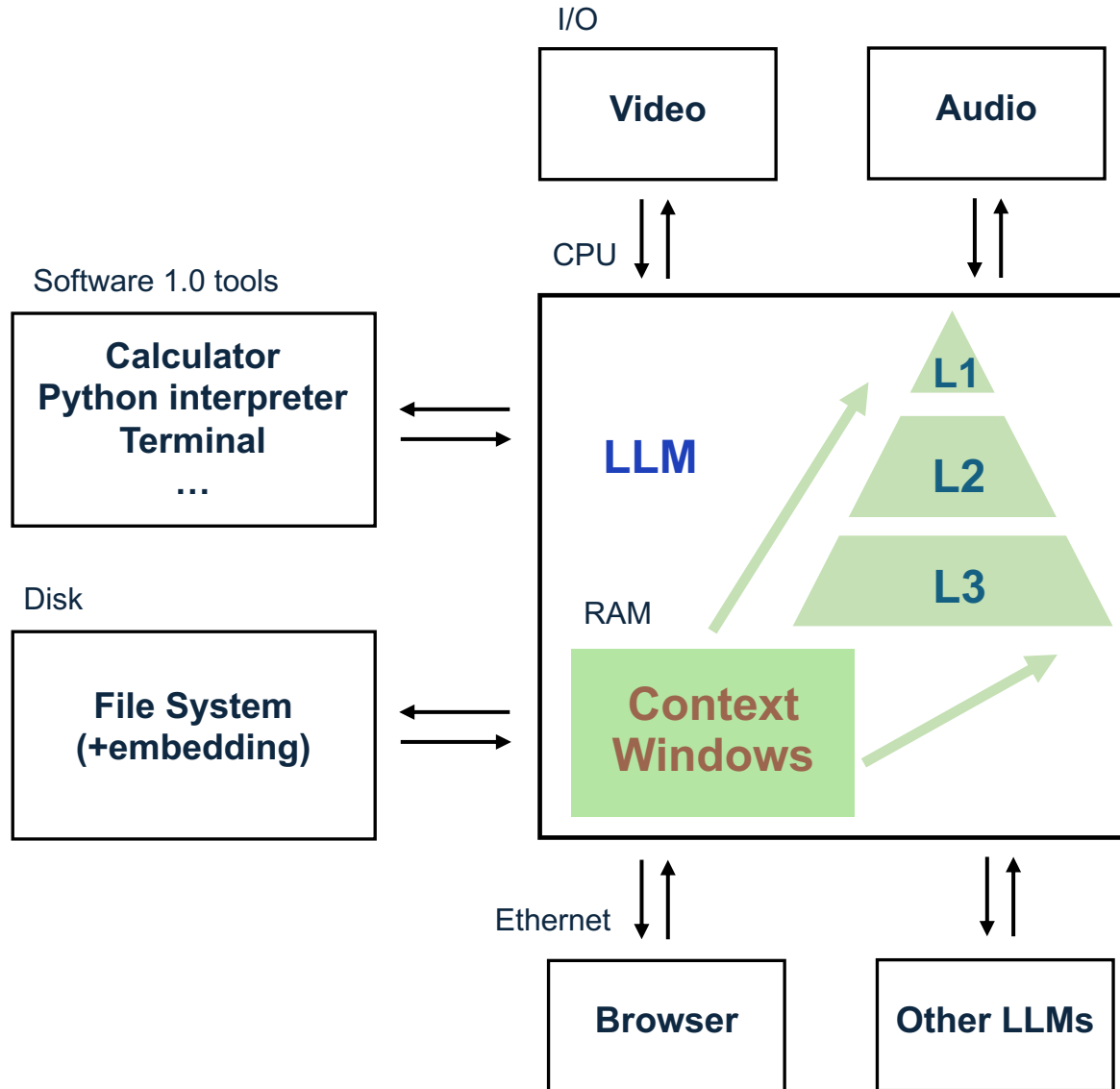


The accuracy in different context windows^[5].

- Different **positions/irrelevant information** can negatively impact the performance of LLMs.

Reorganization Context Within Prompt

Context Windows is RAM of LLM OS



$$\max_{\mathbf{r}, \mathbf{p}} \mathbb{E}_{\mathbf{p}} \sum_{i=1}^K f(\mathbf{r}_i, \mathbf{p}_i)$$

Where $\mathbf{r}_i$ represents the relevance of the i -th document and $\mathbf{p}_i$ denotes the position attentive priority of the i -th document, f is defined as the score function for the i -th document, based on its relevance and position priority.

New requirement for context organization method, analogous to the multi-level cache.

1) Compressing Irrelevant Tokens;

$$(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_k) \xrightarrow{\text{PPL}} (\mathbf{x}_2, \mathbf{x}_7, \dots, \mathbf{x}_{k-3})$$

2) Reorganization Based on Relevance;

$$(\mathbf{x}^{\text{ins}}, \mathbf{x}_1^{\text{doc}}, \dots, \mathbf{x}_{K'}^{\text{doc}}, \mathbf{x}^{\text{que}}) \xrightarrow{r_k} (\mathbf{x}^{\text{ins}}, \mathbf{x}_{r_1}^{\text{doc}}, \dots, \mathbf{x}_{r_{K'}}^{\text{doc}}, \mathbf{x}^{\text{que}})$$

Natural Language is Redundancy

High Information Texts:

- **Speedtalk:** (Gulf, Robert A. Heinlein, 1949)

"*øænn*"

"The far horizons draw no nearer."

- **Traditional Chinese:**

“知不可乎驟得，托遺响于悲風。”

"Life is as fleeting as a mayfly in the vastness of the universe, as insignificant as a grain of rice in the ocean. How I long to be as endless as the Yangtze River, to roam the heavens with immortals, to embrace the bright moon in the highest skies and live eternally. However, I am aware that such dreams are ultimately unattainable. All I can do is transform my regrets and sorrows into the melancholic notes of a flute, entrusting them to the desolate autumn winds."

Low Information Texts:

- **Meeting Recap:**

➤ **John:** So, um, I've been thinking about the project, you know, and I believe we need to, uh, make some changes. I mean, we want the project to succeed, right? So, like, I think we should consider maybe revis(e)ing the timeline.

➤ **Sarah:** I totally agree, John. I mean, we have to be realistic, you know. The timeline is, like, too tight. You know what I mean? We should definitely extend it.

Sapir-Whorf Hypothesis

- Structure of a language influences how its speakers think and perceive the world.

Prompt can be compressed!

LLMs can understand compressed prompt

USER

Compress the following text in a way that fits a Tweet, and such that you (GPT-4) can reconstruct it as close as possible to the original. This is for yourself. Do not make it human readable. Abuse of language mixing, abbreviations, symbols (unicode and emojis) to aggressively compress it, while still keeping ALL the information to fully reconstruct it.

Text to compress:

Lambda calculus may be untyped or typed. In typed lambda calculus, functions can be applied only if they are capable of accepting the given input's "type" of data. Typed lambda calculi are weaker than the untyped lambda calculus, which is the primary subject of this article, in the sense that typed lambda calculi can express less than the untyped calculus can. On the other hand, typed lambda calculi allow more things to be proven. For example, in the simply typed lambda calculus it is a theorem that every evaluation strategy terminates for every simply typed lambda-term, whereas evaluation of untyped lambda-terms need not terminate. One reason there are many different typed lambda calculi has been the desire to do more (of what the untyped calculus can do) without giving up on being able to prove strong theorems about the calculus.

ASSISTANT

LmbC_untyped/typedTypLmbC:funcs_appl✅if
input"type"👉.Typ<untyp(🌀)=exprss👉TypLmbC:prov+.SmplTypLmbC:👉eval_strat_term
👉.untypLmb-term:may_not_term.🔍variousTypLmbC:do+(untyp),👉thrm_provs.

Compress prompt using GPT-4^[6].

USER

I asked you to compress a long text using your own abbreviations. You replied with:

LmbC_untyped/typedTypLmbC:funcs_appl✅if
input"type"👉.Typ<untyp(🌀)=exprss👉TypLmbC:prov+.SmplTypLmbC:👉eval_strat_term
👉.untypLmb-term:may_not_term.🔍variousTypLmbC:do+(untyp),👉thrm_provs.

Reconstruct the original text.

ASSISTANT

Lambda Calculus can be divided into untyped and typed. In the typed lambda calculus, functions can only be applied if the input has the correct "type". Typing prevents some untyped lambda expressions from being expressed, which reduces the risk of logical paradoxes. Typed lambda calculus has been used to provide more powerful type systems. Simple typed lambda calculus, however, does not guarantee the evaluation strategy to terminate. Untyped lambda-term might not terminate, and various typed lambda calculi have been developed to do more (than untyped) and provide stronger guarantees for theorem provers.

Recovering compressed prompt in another session.

Compressed prompt is useful!

* [6] <https://twitter.com/VictorTaelin/status/1642664054912155648>

(Long)LLMLingua: Accelerating and Enhancing LLM Inference

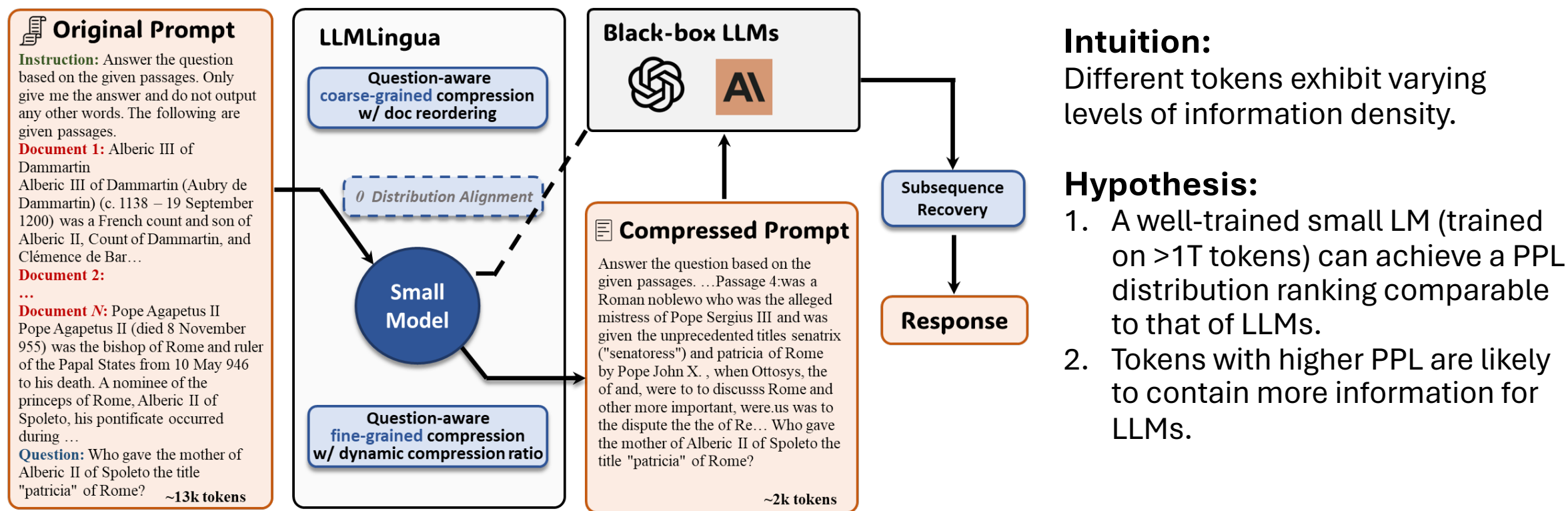
Design | Impact | Batch Evaluation | Cases Studies

* [7] LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models.

[8] LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression.

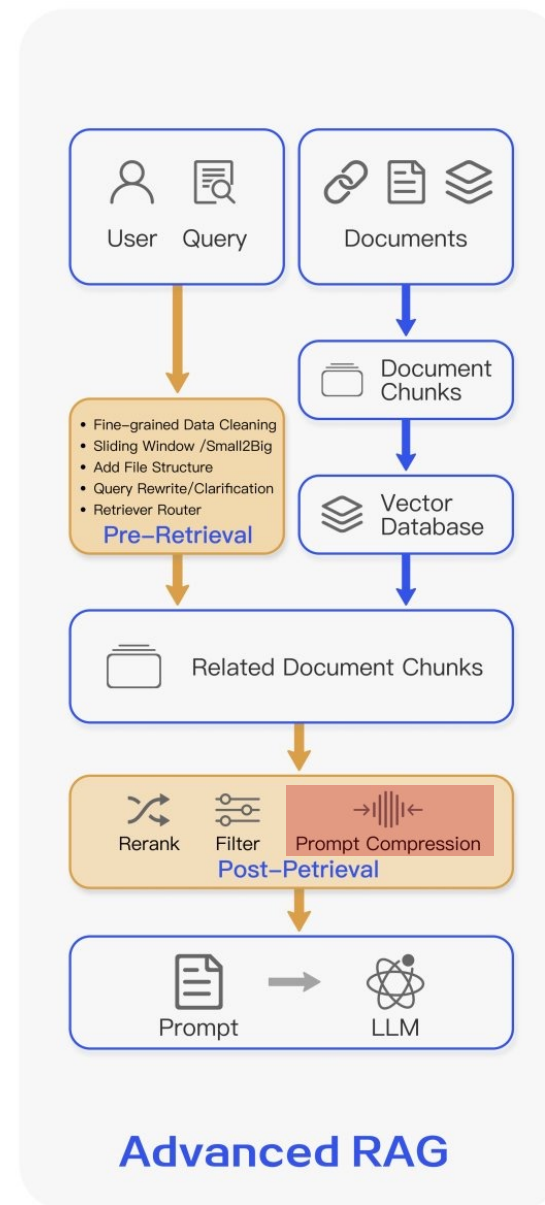
Design Overview

(Long)LLMLingua: a coarse-to-fine question-aware prompt compression framework to enhance LLMs' perception of key Information.



Impact

- 🏆 Some Hackathon Award.
- 🤖 Used by [LlamaIndex](#), which is a widely used RAG framework.
- 🤖 Summarized as an important module in Advanced RAG.



Question/Task-Aware Coarse-to-Fine Compression

Eliminating irrelevant information in prompts to boost LLMs' perception of key information.

❶ Coarse-Grained compression at doc level.

$$r_k = \frac{1}{N_c} \sum_i^{N_c} p(x_i^{\text{que}, \text{restrict}} | \mathbf{x}_k^{\text{doc}}) \log p(x_i^{\text{que}, \text{restrict}} | \mathbf{x}_k^{\text{doc}})$$


Restrictive statement: “We can get the answer to this question in the given documents”.

To strengthen the interconnection between the question x^{que} and the doc x_k^{doc} .

Computation: 1 extra inference on whole prompt, plus K extra inferences on $x^{\text{que}, \text{restrict}}$ (acceptable).

Reorder documents using their relevance score to mitigate LLMs' position bias.

$$(\mathbf{x}^{\text{ins}}, \mathbf{x}_1^{\text{doc}}, \dots, \mathbf{x}_{K'}^{\text{doc}}, \mathbf{x}^{\text{que}}) \xrightarrow{r_k} (\mathbf{x}^{\text{ins}}, \mathbf{x}_{r1}^{\text{doc}}, \dots, \mathbf{x}_{rK'}^{\text{doc}}, \mathbf{x}^{\text{que}})$$

Original prompt: $\mathbf{x} = \{x^{\text{ins}}, x_1^{\text{doc}}, \dots, x_{K'}^{\text{doc}}, x^{\text{que}}\}$

x^{ins} can be removed, x_k^{doc} can be any additional materials appended by users, and x^{que} can be any requirement specified by users.

❷ Fine-grained compression at token level.

$$s_i = \text{perplexity}(x_i | x_{<i}) - \text{perplexity}(x_i | x^{\text{que}}, x_{<i})$$

Contrastive perplexity: our mathematical analysis shows that $s_i \propto p(x^{\text{que}} | x_i, x_{<i})$. But More Efficient!

Computation: <2 extra inference on whole prompt (acceptable).

Iterative compression with dynamic compression ratio for better dependency modeling and adaptive granular control.

$$\tau_i = \tau_k^{\text{doc}}, \quad \forall x_i \in \mathbf{x}_k^{\text{doc}},$$
$$\tau_k^{\text{doc}} = \max(\min((1 - \frac{2I(r_k)}{K'})\delta\tau + \tau^{\text{doc}}, 0), 1)$$

Empirical Evidence of Question/Task-Aware Compression

❶ Coarse-Grained compression at doc level.

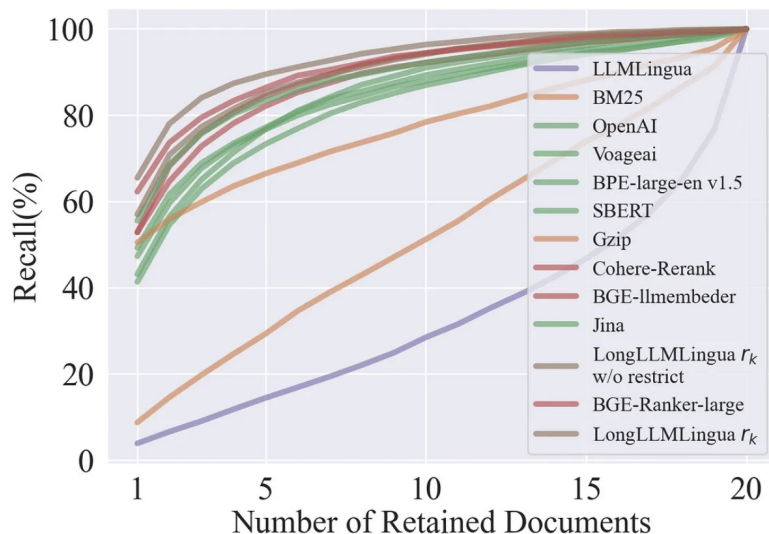
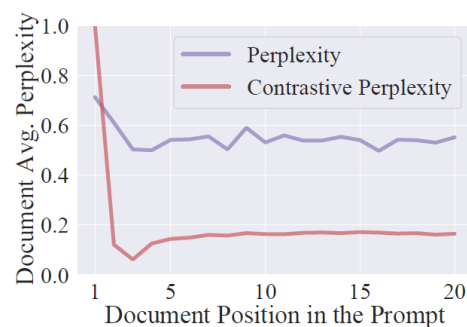
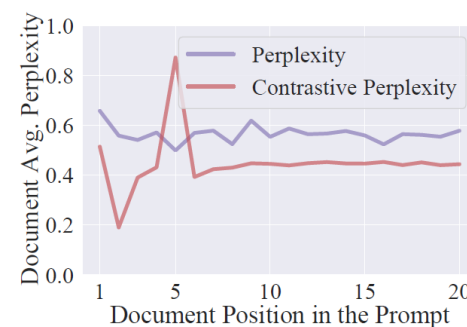


Figure 1. The accuracy of different methods for ranking documents from Multi-document QA dataset, which increases from top to bottom in terms of Recall@1. Different colors represent different types of methods. Among them, yellow represents traditional relevance methods, green signifies embedding-based methods, and red denotes rerank-based methods. You can find the script in [this link](#).

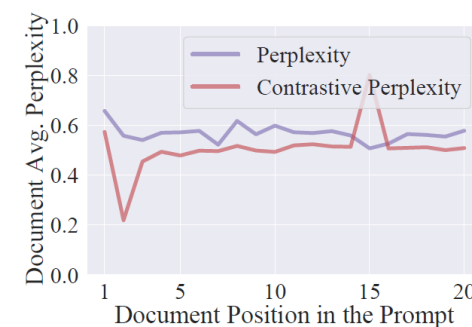
❷ Fine-grained compression at token level.



(a) 1st



(b) 5th



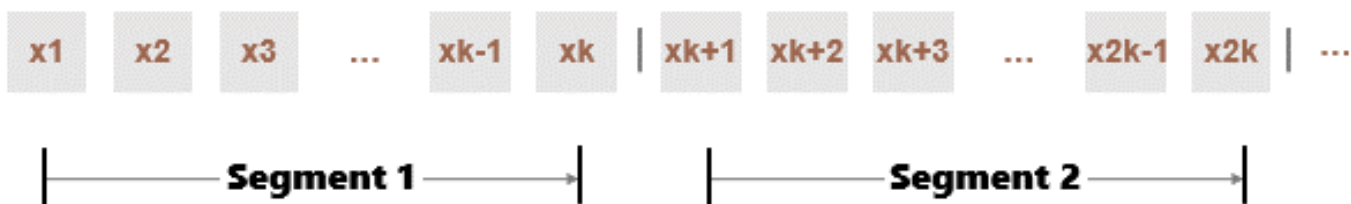
(c) 15th

Figure 5: The distribution of document-level average perplexity when the ground-truth document is in different positions.

$$\begin{aligned}
 s_i &= \text{perplexity}(x_i | x_{<i}) - \text{perplexity}(x_i | x^{\text{que}}, x_{<i}) \\
 &= q(x_i) \log p(x_i | x^{\text{que}}, x_{<i}) - q(x_i) \log p(x_i | x_{<i}) \\
 &= q(x_i) \log \frac{p(x_i | x^{\text{que}}, x_{<i})}{p(x_i | x_{<i})} \propto p(x^{\text{que}} | x_i, x_{<i})
 \end{aligned}$$

KV Cache in Fine-Grained Iterative Prompt Compression

Original Prompt



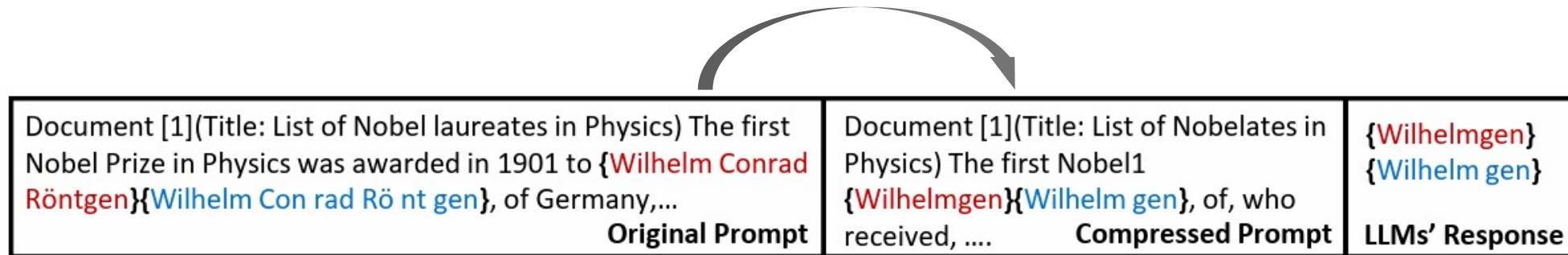
$$s_i = \text{perplexity}(x_i | x_{<i}) - \text{perplexity}(x_i | x^{\text{que}}, x_{<i})$$

Contrastive perplexity: our mathematical analysis shows that $s_i \propto p(x^{\text{que}} | x_i, x_{<i})$. But More Efficient!

Post-compression Subsequence Recovery

To improve the integrity of key information.

Token-level compression may drop key entity tokens.



Blue texts: tokenized results.

String matching



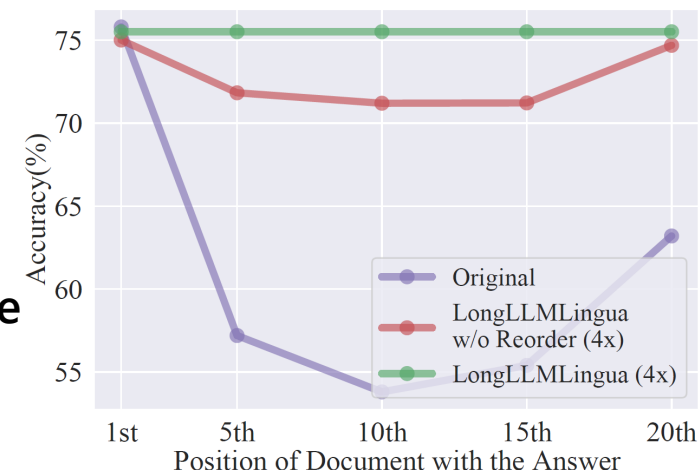
Wilhelm Conrad Röntgen

Evaluation on QA

Methods	GPT3.5-Turbo						Length		Latency	
	1st	5th	10th	15th	20th	Reorder	Tokens	$1/\tau$	Latency	Speedup
<i>2x constraint</i>										
<i>Retrieval-based Methods</i>										
BM25	53.7	49.3	47.9	49.9	46.9	50.3	1,545	1.9x	2.1	1.9x
Gzip	64.6	63.8	60.5	58.3	57.3	64.4	1,567	1.9x	2.1	1.9x
SBERT	72.5	67.9	63.3	65.0	66.2	68.7	1,549	1.9x	2.2	1.9x
OpenAI	73.0	65.6	66.5	65.4	65.5	69.9	1,550	1.9x	4.9	0.8x
LLMLingua r_k	73.9	67.7	68.7	66.0	65.6	74.3	1,548	1.9x	2.3	1.8x
<i>Compression-based Methods</i>										
Selective-Context	45.4	39.0	33.8	33.5	41.5	-	1,478	2.0x	7.4	0.6x
LLMLingua	77.2	72.9	70.8	70.5	70.6	76.2	1,429	2.1x	2.9	1.4x
<i>4x constraint</i>										
<i>Retrieval-based Methods</i>										
BM25	40.6	38.6	38.2	37.4	36.6	36.3	798	3.7x	1.5	2.7x
Gzip	63.1	61.0	59.8	61.1	60.1	62.3	824	3.6x	1.5	2.7x
SBERT	66.9	61.1	59.0	61.2	60.3	64.4	808	3.6x	1.6	2.5x
OpenAI	63.8	64.6	65.4	64.1	63.7	63.7	804	3.7x	4.3	1.0x
LLMLingua r_k	71.1	70.7	69.3	68.7	68.5	71.5	807	3.7x	1.7	2.4x
<i>Compression-based Methods</i>										
Selective-Context	31.4	19.5	24.7	24.1	43.8	-	791	3.7x	6.8	0.6x
LLMLingua	75.0	71.8	71.2	71.2	74.7	75.5	748	3.9x	2.1	2.0x
Original Prompt	75.7	57.3	54.1	55.4	63.1	-	2,946	-	4.1	-
Zero-shot	56.1						15	196x	1.1	3.7x

Evaluation on NaturalQuestions.

Mitigate “Lost in the Middle” Issue and Boost performance.



LLMLingua gains a **performance boost of up to 21.4%** over the original prompt with **~4x fewer** tokens as input to GPT-3.5-Turbo.

Speedup Latency.

LLMLingua can **speed up the end-to-end latency** by up to **2.0x** under the 4x compression ratio.

Save cost.

LLMLingua can drive **cost savings up to \$93.6** per **1,000 samples** from the LooGLE benchmark.

	Multi-document QA	LongBench	ZeroScrolls	MuSicQue	LooGLE
Original	4.6	31.5	30.6	3.8	93.6
Ours	1.3	3.0	3.2	1.8	5.6

Evaluation on Math & Reasoning

Similar conclusions are derived. Compressed prompts can effectively retain reasoning information.

Methods	GSM8K			BBH		
	EM	Tokens	1/ τ	EM	Tokens	1/ τ
Full-shot	78.85	2366	-	70.07	774	-
1-shot constraint						
Sentence Selection	77.10	422	6x	69.60	284	3x
Selective-Context	53.98	452	5x	54.27	276	3x
GPT4 Generation	71.87	496	5x	27.13	260	3x
Ours	79.08	439	5x	70.11	288	3x
half-shot constraint						
Sentence Selection	72.33	230	10x	39.56	175	4x
Selective-Context	52.99	218	11x	54.02	155	5x
GPT4 Generation	68.61	223	11x	27.09	161	5x
Ours	77.41	171	14x	61.60	171	5x
quarter-shot constraint						
Sentence Selection	66.67	195	12x	46.00	109	7x
Selective-Context	44.20	157	15x	47.37	108	7x
GPT4 Generation	56.33	188	20x	26.81	101	8x
Ours	77.33	117	20x	56.85	110	7x
zero-shot	25.78	11	215x	32.32	16	48x
Simple Prompt	74.9	691	3x	-	-	-

8-shot 3-step CoT.

https://github.com/FranxYao/chain-of-thought-hub/blob/main/gsm8k/lib_prompt/prompt_easy.txt

$1/\tau$	1x	2x	5x	10x
End-to-End w/o LLMingua	8.6	-	-	-
End-to-End w/ LLMingua	-	4.9 (1.7x)	2.3 (3.3x)	1.3 (5.7x)
LLMingua	-	0.8	0.3	0.2

Table E. Latency (s) comparison on GSM8K.

Approach the score of 62.0 achieved by full-shot PaLM-540B.

More tasks...

Methods	LongBench										ZeroSCROLLS			
	SingleDoc	MultiDoc	Summ.	FewShot	Synth.	Code	AVG	Tokens	1/ τ	Latency	AVG	Tokens	1/ τ	Latency
<i>3,000 tokens constraint</i>														
<i>Retrieval-based Methods</i>														
BM25	32.3	34.3	25.3	57.9	45.1	48.9	40.6	3,417	3x	7.5(2.1x)	19.8	3,379	3x	5.5(2.2x)
SBERT	35.3	37.4	26.7	63.4	51.0	34.5	41.4	3,399	3x	7.7(2.0x)	24.0	3,340	3x	5.9(2.1x)
OpenAI	34.5	38.6	26.8	63.4	49.6	37.6	41.7	3,421	3x	13.3(1.2x)	22.4	3,362	3x	11.7(1.0x)
LLMLingua r_k	37.6	42.9	26.9	68.2	49.9	53.4	46.5	3,424	3x	8.2(1.9x)	29.3	3,350	3x	6.2(2.0x)
<i>Compression-based Methods</i>														
Selective-Context	23.3	39.2	25.0	23.8	27.5	53.1	32.0	3,328	3x	50.6(0.3x)	20.7	3,460	3x	54.2(0.2x)
LLMLingua r_k	40.7	46.2	27.2	70.6	53.0	55.2	48.8	3,283	3x	8.0(1.6x)	32.8	3,412	3x	8.2(1.5x)
<i>2,000 tokens constraint</i>														
<i>Retrieval-based Methods</i>														
BM25	30.1	29.4	21.2	19.5	12.4	29.1	23.6	1,985	5x	4.6(3.4x)	20.1	1,799	5x	3.8(3.2x)
SBERT	33.8	35.9	25.9	23.5	18.0	17.8	25.8	1,947	5x	4.8(3.4x)	20.5	1,773	6x	4.1(3.0x)
OpenAI	34.3	36.3	24.7	32.4	26.3	24.8	29.8	1,991	5x	10.4(1.5x)	20.6	1,784	5x	9.9(1.2x)
LLMLingua r_k	37.8	41.7	26.9	66.3	53.0	52.4	46.3	1,960	5x	4.7(3.3x)	24.9	1,771	6x	10.4(1.2x)
<i>Compression-based Methods</i>														
Selective-Context	16.2	34.8	24.4	15.7	8.4	49.2	24.8	1,925	5x	47.1(0.3x)	19.4	1,865	5x	47.5(0.3x)
LLMLingua r_k	39.0	42.2	27.4	69.3	53.8	56.6	48.0	1,809	6x	6.1(2.6x)	32.5	1,753	6x	5.2(2.3x)
Original Prompt	39.7	38.7	26.5	67.0	37.8	54.2	44.0	10,295	-	15.6	32.5	9,788	-	12.2
Zero-shot	15.6	31.3	15.6	40.7	1.6	36.2	23.5	214	48x	1.6(9.8x)	10.8	32	306x	1.0(12.2x)

LLMLingua generalizes well to other (L/S)LMs.

- Similar conclusions have been observed when using:
 - **Small LM for compression:** from [Alpaca-7B](#) to [GPT2-small-Alpaca](#).
 - **LLMs:** from [GPT-3.5-turbo](#) to [LongChat-13b](#) and [Claude-v1.3](#).

GPT2-small-Alpaca on GSM8k	EM	Tokens	1/ τ
Ours with GPT2 in 1-shot constraint	77.02	447	5x
Ours with GPT2 in half-shot constraint	76.42	173	14x
Ours with GPT2 in quarter-shot constraint	76.27	128	18x

Claude-v1.3 on GSM8k	EM	Tokens	1/ τ
Ours in 1-shot constraint	83.51	439	5x
Ours in half-shot constraint	82.61	171	14x
Simple Prompt	81.8	691	3x

Methods	LongChat-13b						Length	
	1st	5th	10th	15th	20th	Reorder	Tokens	1/ τ
<i>2x constraint</i>								
<i>Retrieval-based Methods</i>								
BM25	50.9	44.9	44.1	42.9	43.2	46.0	1,545	1.9x
Gzip	61.9	55.7	52.7	50.8	50.9	59.3	1,567	1.9x
SBERT	65.8	57.5	54.9	53.4	55.7	61.4	1,549	1.9x
OpenAI	65.9	57.5	56.2	54.2	55.7	61.7	1,550	1.9x
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Selective-Context	53.2	26.3	25.4	24.2	33.3	-	1,478	2.0x
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<i>4x constraint</i>								
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Gzip	57.6	52.9	51.0	50.1	50.4	57.2	824	3.6x
SBERT	62.6	56.6	55.1	53.9	55.0	59.1	808	3.6x
OpenAI	61.2	56.0	55.1	54.4	55.0	58.8	804	3.7x
LongLLMLingua r_k	67.8	59.4	57.7	57.7	58.6	64.0	807	3.7x
<i>Compression-based Methods</i>								
Selective-Context	38.2	17.2	15.9	16.0	27.3	-	791	3.7x
LLMLingua	68.7	60.5	59.3	58.3	61.3	66.7	748	3.9x
Original Prompt	68.6	57.4	55.3	52.5	55.0	-	2,946	-
Zero-shot							15	196x

GPT-4 can recovery the compressed prompt.

Original Prompt(9-steps Chain-of-Thought)/Compressed Prompt:

Question: Sam bought a dozen boxes, each with 30 highlighter pens inside, for \$10 each box. He rearranged five of these boxes into packages of six highlighters each and sold them for \$3 per package. He sold the rest of the highlighters separately at the rate of three pens for \$2. How much profit did he make in total, in dollars?

Let's think step by step

- 1 Sam bought 12 boxes $\times$ \$10 = \$120 worth of highlighters.
- 2 He bought $12 \times 30 = 360$ highlighters in total.
- 3 Sam then took 5 boxes $\times$ 6 highlighters/box = 30 highlighters.
- 4 He sold these boxes for $5 \times \$3 = \15
- 5 After selling these 5 boxes there were $360 - 30 = 330$ highlighters remaining.
- 6 These form $330 / 3 = 110$ groups of three pens.
- 7 He sold each of these groups for \$2 each, so made $110 \times 2 = \$220$ from them.
- 8 In total, then, he earned $\$220 + \$15 = \$235$.
- 9 Since his original cost was \$120, he earned $\$235 - \$120 = \$115$ in profit.

The answer is 115

1. Abbreviations and Short Forms

- "highlighter" -> "highl"
- "rearranged" -> "reanged"
- "highlighters" -> "thelsters"

2. Numerical Conversions

- " $\times$ \$10" -> " $\times 0$ "
- " 12×30 " -> " $2\ 3$ "

3. Omission of Words or Syllables

- "each with 30 highlighter pens inside" -> "each 30 highl pens inside"
- "packages of six highlighters each and sold them for \$3 per package" -> "of six each \$3 per"

4. Eliminating Redundant Words or Phrases

- "Let's think step by step" -> "Lets think step"
- "Sam bought 12 boxes $\times$ \$10 = \$120 worth of highlighters" -> "bought boxes $\times 0$ oflters"

Recovery Prompt (9-steps Chain-of-Thought):

Question: Sam purchased a dozen boxes, each containing 30 highlighter pens, for \$10 each. He rearranged five of these boxes to have six pens each and sold them for \$3 per box. He then sold the remaining highlighters individually in groups of three for \$2 each. Let's calculate the total profit.

2 Initially, Sam bought 12 boxes with 30 highlighters each, which equals 360 highlighters.

1 He spent \$10 per box, so his total cost was \$120.

3 Sam then rearranged 5 boxes to have 6 highlighters per box, which equals 30 highlighters.

4 He sold these boxes for \$3 each, making \$15.

5 After selling these boxes, he had 330 highlighters left ($360 - 30$).

6 He sold these in groups of three, which equals 110 groups.

7 He sold each group for \$2, making \$220 from this.

8 In total, Sam made \$235 ($\$15 + \220).

9 Since his initial cost was \$120, he earned a profit of \$115 ($\$235 - \120).

The answer is \$115.

Cases from real-world scenarios-RAG

Original Prompt:

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

.....
Document [2](Title: OPEC) of "the top 100 most influential people in the shipping industry". However, the influence of OPEC on international trade is periodically challenged by the expansion of non-OPEC energy sources, and by the recurring temptation for individual OPEC countries to exceed production targets and pursue conflicting self-interests. As of June 2018, OPEC has 15 member countries: six in the Middle East (Western Asia), seven in Africa, and two in South America. According to the U.S. Energy Information Administration (EIA), OPEC's combined rate of oil production (including gas condensate) represented 44 percent of the world's total in 2016, and OPEC accounted for

.....
Document [10](Title: OPEC) Organization of the Petroleum Exporting Countries (OPEC, /'oʊpek/ OH-pek, or OPEP in several other languages) is an intergovernmental organization of 14 nations as of February 2018, founded in 1960 in Baghdad by the first five members (Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela), and headquartered since 1965 in Vienna, Austria. As of 2016, the 14 countries accounted for an estimated 44 percent of global oil production and 73 percent of the world's "proven" oil reserves, giving OPEC a major influence on global oil prices that were previously determined by American-dominated multinational oil companies.

.....
Question: how many countries are a part of opec in may 2018

Answer:

Compressed Prompt:

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

[5Title: OPEC OPE its two newest, who had joined mid1970s. Ecuador withdrew in December 1992 it was unwilling pay US\$2 million membership fee and felt that it needed to produce more it was allowed under the OPEC quota, although rejoined in October 2007. Similar concerns prompted Gabon to suspend membership in January 1995; it rejoined in July 2016 Iraq remained a member of OPEC since the organization's founding, Ira production was a part of OPEC quota agreements from 1998 to 216, due to the country's daunting political difficulties.

1 the Petroleum Exporting Countries (OPEC, /'oʊpek/ OH-pek, or OPEP in several other languages) is an intergovernmental organization of 14 nations as of February 2018, founded in 1960 in Baghdad by the first five members (Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela), and headquartered since 1965 in Vienna, Austria. As of 2016, the 14 countries accounted for an estimated 44 percent of global oil production and 73 percent of the world's "proven" oil reserves, giving OPEC a major influence on global oil prices that were previously determined by American-dominated multinational oil companies.

Question: how many countries are a part of opec in may 2018

Answer:

Tokens 3k -> 313 (9X), Acc 54.1 -> 75.5 (+21.4)

Cases from real-world scenarios-Online Meeting

Original Prompt:

Speaker 4: We're going to hear all of the presentations at once. And then after we go through all the presentations, we'll do all the all of the questions at once and then any any public comment, and we'll go from there. Thank you.

Speaker 7: All right. Thank you, Mr. Mayor. It was our pleasure to give you an overview of the budget last Tuesday, we transmitted that officially to the mayor and city council and the public. And now we're looking to really dove deeper in some of the key departments. Today, we want to go through three key departments, the fire department. Then we'll go through our police department and then Parks, Recreation and Marine. So I will turn it over to Chief Espino and his team to go through the fire department presentation first followed by police, followed by Parks and Rec.

Speaker 1: It's.

Speaker 6: Thank you, Mr. Monaco. Good evening. Mayor Garcetti and members of the City Council. This evening, I will be providing a brief overview of the fire department, highlighting the key services that we provide to the community and some of our recent accomplishments.Over the past 12 months, we've responded to over 80,000 fire, emergency medical services, marine safety and other emergency incidents, a 5% increase from last year.

.....

Speaker 9: Thank you, Mr. Modica. Ms.. Tatum. Good afternoon, Honorable Mayor and members of the City Council. I'm very excited to be presenting the Police Department's

.....

Question: How much did the crime rate increase last year?

Answer:

Compressed Prompt:

aker3., the.

Speaker 9 : Thank youoman Yes,'s. 5.4% increase to date That after this a 1.4 increase in crime in. : you Let keep police:: Investig : No, Counciloman, it is the federal UCR one violentized onines. can say the gets day. Our department will continue to evolve and move forward, building on our existing strengths and taking advantage of opportunities for growth and renewal. Our priorities around crime and homelessness, employee and community wellness and open communication will help guide us further into 21st century policing, while also supporting the shared responsibility of public safety in the city of Long Beach. Thank you. Myself and Bureau Chief Josie Murray stand ready to answer any questions they can.

Question: How much did the crime rate increase last year?

Answer:

Tokens 30k -> 172 (175X)

You can find more real-world cases in

aka.ms/LLMLingua

Takeaway

- As prompts grow longer, they often entail higher costs and **struggle to deliver information effectively** to LLMs.

- => 1. Longer prompts tend to contain more **noise**;
2. LLMs have **position attentive priority**;

=> So **new requirement** to **reorganization** the context within prompt;

- We propose **(Long)LLMLingua**,

1. **Compression irrelevant tokens**;

Insight #1: Language is redundancy.

#2: LLMs can understand compressed prompt.

2. **Reorganization based on relevant**;

Insight #3: LLMs have position attentive priority.

Our method can **mitigate “Lost in the Middle” issue** (improve 21.4% only using ¼ tokens) and saving up to \$93.6 per 1,000 examples.